

Preliminary Community Assessment of Reproducibility Status for Claims of Suppressed Fatigue in Hierarchical Alloys

T. Johansson^{1,*}, K. Roy², M. Spengler¹, and J. Carvalho³

¹*Department of Mechanical Engineering, Gothenburg Technical Institute, Gothenburg, Sweden*

²*Materials Reproducibility Initiative, Cornell University, Ithaca, NY, USA*

³*Laboratório de Fratura e Fadiga, Universidade de São Paulo, São Paulo, Brazil*

*t.johansson@gti.se

ABSTRACT

We report the status of an international, informally coordinated effort to evaluate the reproducibility of the claim by Kawabata et al. that hierarchical metallic architectures exhibit no measurable fatigue degradation up to 10^{12} cycles. We surveyed 47 laboratories with fatigue testing capability, of which 18 indicated active or planned replication, 21 indicated intent to wait for methodology clarification, and 8 declined to engage. We summarize the principal methodological concerns raised in the community discussion, map them to the relevant experimental steps, and outline a consensus minimum replication protocol that distributes verification across independent laboratories. We emphasize that this is a preliminary report intended to coordinate effort and prevent duplication, not to adjudicate the underlying claim.

1 Background

Within the first 72 hours following posting of the preprint by Kawabata et al. [1], online discussion across technical and community forums reflected a bimodal response: substantive engagement with the methodology on one hand, and skepticism rooted in the long historical reference class of overstated materials claims on the other. The concurrent letter by Zhang et al. [2] partially addressed the "single laboratory artifact" concern but did not resolve the broader question of reproducibility in independent hands.

We describe the results of a structured outreach to laboratories worldwide, conducted between days 3 and 14 after preprint posting. The purpose of this paper is twofold: first, to reduce duplicated effort and gaps in replication coverage by publishing a registry of committed activity; second, to codify a minimum common protocol.

2 Methodology

2.1 Survey

We identified 63 research groups globally with published fatigue-testing capability at least at 10^7 cycles and reachable institutional contact information. 47 of the 63 responded within the survey window. Responses were categorized as: *active* (replication begun), *planned* (within 6 months), *waiting* (pending methodology clarification), or *declining*.

2.2 Consensus protocol development

We convened an open online workshop with 34 participants drawn from respondent laboratories. Participants were asked to enumerate the minimum set of measurements re-

quired to either (a) reproduce the original claim or (b) falsify it convincingly. A consensus protocol was drafted by rotating co-authors and circulated for 5 days of comment.

3 Findings

3.1 Current replication landscape

Table 1 summarizes the distribution of committed effort across geographies and capability classes. Three capability gaps were identified: (i) only two of the responding laboratories currently have piezoelectric resonant fatigue systems capable of reaching 10^{12} cycles within 2 years; (ii) only one laboratory has combined *in-situ* TEM fatigue capability at the required displacement resolution; (iii) no respondent reported capability for multiaxial fatigue at these cycle counts.

Table 1: Survey response summary.

Category	Count	Fraction
Active replication	18	38%
Planned replication	14	30%
Awaiting clarification	9	19%
Declined	6	13%
Total responding	47	100%

3.2 Principal methodological concerns

Respondents identified the following concerns most frequently:

1. The claim that cellular substructures of 0.3 to 1.0 μm

(12 to 40 μin) from LPBF processing necessarily reduce fatigue life is inconsistent with some recent results. The reference-class argument in the original preprint may therefore be weaker than claimed (7 of 18 active laboratories).

2. Forced-air cooling sufficient to hold specimen temperature rise below 3 °C (5.4 °F) at 20 kHz loading is not straightforward for specimens of the reported cross section. Thermal instrumentation details are requested (6 of 18).
3. The particular ISFA parameters, elastic strain of 0.15%, 780 °C (1436 °F), 45 min, are a narrow window and the sensitivity to these parameters has not been characterized (11 of 18).
4. No failure case is reported in the original work. Absence of failure cases complicates assessment of the stochastic component of the mechanism (5 of 18).

3.3 Proposed minimum protocol

We propose that any claimed replication include, at minimum: (i) full processing parameter disclosure including laser scan path; (ii) independent chemical analysis of the powder feedstock; (iii) parallel fatigue testing at two frequencies differing by at least a factor of 10; (iv) interrupted testing with metallographic and TEM inspection at 10^6 , 10^8 , and 10^{10} cycles; and (v) blind comparison of at least two specimen groups processed at independent facilities.

4 Discussion

We explicitly do not draw conclusions about the underlying claim. The state of evidence is consistent with both (a) a genuine mechanism with a narrow processing window, and (b) a measurement artifact compounded by favorable material selection. The purpose of coordinated replication is to distinguish these hypotheses as efficiently as possible. We estimate that the minimum time for a conclusive community assessment is 18 to 24 months.

References

- [1] Y. Kawabata et al., *arXiv* 2604.11891 (2026).
- [2] L. Zhang et al., *arXiv* 2604.12207 (2026).
- [3] A. Ito, M. Grzegorzewski, P. Delacroix, *arXiv* 2604.12884 (2026).